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Purpose/Policy:

“This “Keg Rebuilding” standard operating procedure has been created to assist operators with setup, operations, changeover, cleaning, and troubleshooting.

Scope:

The purpose of the “Keg Rebuilding” is to fix kegs that have been taken out of circulation due to missing parts or flaws. In the instance the keg is verified to be damaged beyond repair, the keg becomes scrap.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- CO2 – Carbon Dioxide
- Spear – The valve assembly in the neck of a keg
- Coupler – Fitting that attaches to the keg spear to move liquid or gas in and out of the keg

Authorities/Responsibilities:

“This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Keg line packaging team. The objective of this SOP is to gain the knowledge needed to complete the “Keg Rebuilding SOP” and trouble shoot any issues that may arise.”

Resources and Materials:

Operations:

- Removal and replacement of parts in a safe

Changeovers:

- Not Applicable

Basic Cleaning:

- Ensure clean rebuild/new parts are being added to rebuild kegs.

Deep Clean:

- N/A.

Troubleshooting:

- No Equipment required.

PPE Required:

- Safety Glasses are required.

Procedures:

Verifying Keg Condition:

1. Inspect the keg that is staged for rebuilding to see if the top or bottom of the keg is blown out. The keg should have its manufacturing stamp present.
   1.1 Please see the following figures for “Good” keg conditions:

Figure 1: Top of Good Keg



Figure 2: Botom of Good Keg



Figure 3: Side profile of a good keg. Note the even, shallow bottom dome, with the stampings intact.

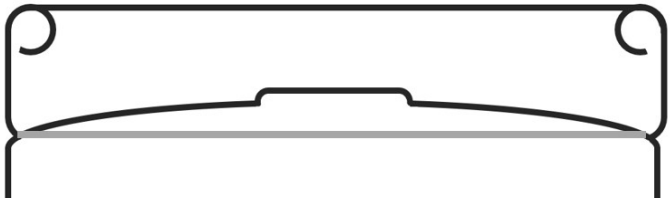


Figure 4: Example of a Good Keg Neck (ensuring it is straight with no bends or cracks)



1.2 Please see the following figures for “Bad” keg conditions.

Figure 5: Bad Keg (Blown out): Side profile of a blown (bad) keg. Look for a “bloated” appearance or signs of rupture. These are to be scrapped. The bottom dome will frequently extend past the bottom chime and the keg will not sit flat – this is a dead giveaway.

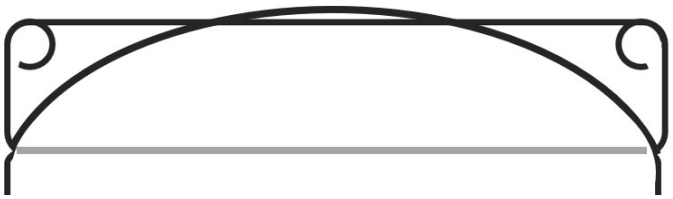


Figure 6: Bad Keg (Bent Neck): Can be possible fixed, see Figure 7A and 7B and instructions.



1.3 There may be a chance that a bent neck is fixable. To do so, follow the steps below:

1.3.1 To straighten the neck, install the straightening bar into the coupler.

Figure 7A: Straightening Bar



1.3.2 Install the leverage bar onto the straightening bar. Bend the neck back into position.

Figure 7B: Straightening Bar with Leverage Bar



1.3.3 If the neck seems soft or moves back into the bent position the keg is no good and should be scrapped.

Depressurizing Kegs:

1. To depressurize the keg, you will need a “D” system coupler with a drain hose.

Figure 8: “D” System Coupler



2. Insert the drain hose into the floor drain located on the Keg Line.
3. Insert and twist the “D” system coupler until it locks in place.
4. Press down on the coupler’s lever and lock it into place.
    4.1 This will release all pressure inside of the keg as well as any residual beer.

Figure 9: “D” System Coupler in the plunged position.



Note: Before proceeding to the next steps, ensure that the keg has been fully relieved of all pressure and any remaining product. Please see a Senior Level Operator or Supervisor for proper procedure verification.

Removing the Keg Spear:

1. As stated in the note above, verify there is no pressure or product left in the keg as trained by a Senior Level Operator or Supervisor.
2. To verify the keg has been depressurized, press down on the “Ball” of the coupler. No air pressure or hissing should be present.

Figure 10: Pressing “Ball” of Coupler



Removing the Locking Ring:

1. First, locate the end of the locking ring.

Figure 11: End of locking ring.



2. Install the “Lock Ring Removal Tool” onto the coupler. Making sure the end of the locking rind is lined up with the “hole” in the removal tool.

Figure 12: Locking tool installed on keg.



3. With the tool in place, turn the internal handle clockwise until it stops.
4. As the handle turns, the locking ring should start to appear through the hole.

Figure 13: Locking Ring poking through hole.



5. With the end of the locking ring showing through the hole, continue to turn the removal tool clockwise until the entire ring has been removed.

Figure 14: Locking ring partially through.



Figure 15: Locking ring all the way through.



Removing the Spear:

1. Turn the spear so the “ears” line up with the “ear channels.”

Figure 15: Spear lined up.



2. The spear should now be removable. Throw the old locking ring away.

Rebuilding the Spear:

1. Grab the spear rebuilding tool.

Figure 16: Rebuilding tool



2. Insert the spear into the rebuilding tool.

Figure 17: Spear on Rebuilding tool.



3. Press down on the spear using the palm of your hand and turn clockwise.

Figure 18: Spear turning.



4. The coupler can now be removed from the spear.
5. Remove the O-ring from the coupler and discard.

Figure 19: Coupler Disassembled.



6. With the Spear on the “Spear rebuilding tool”, remove the CO2 valve gasket.
    6.1 Inset a “hook tool” between the ball and the gasket and pull the gasket up.

Figure 20: Removing CO2 valve gasket:



7. Install a new CO2 gasket onto the ball and spear.

Figure 21: New CO2 valve gasket installed



8. Replace the O-Ring on the coupler.

Figure 22: Replacing O-Ring.



9. Reassemble the spring and then the locking ring with the small tabs facing up on the spear and coupler.

Figure 23: Spring rebuilt.



10. Place the coupler back onto the spear, press down and turn clockwise.

Figure 24: Coupler replacement.



11. The ears on the bottom of the coupler should click into the locking ring.

Figure 25: Rebuilt Spear.



12. Now that the spear has been rebuilt, it can be reinserted into the keg.

Reinserting the Spear:

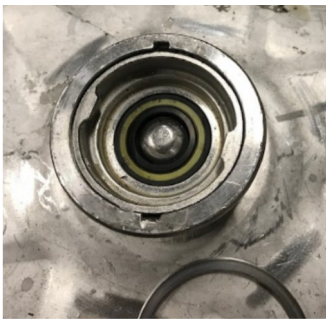
1. Put the spear back into the keg lining up the ears with the ear channels.

Figure 26: Spear being placed into the keg.



2. Once in place, turn the spear clockwise to lock the ears in place.

Figure 27: Ears locked in place



3. Place the new locking ring onto the spear compression tool.

Figure 28: Locking ring on the compression tool.



4. Place the compression tool onto the neck of the keg. Tighten down the tool making sure the ears of the coupler go into the channels of the compression tool.

Figure 29: Compression tool on the neck of the keg



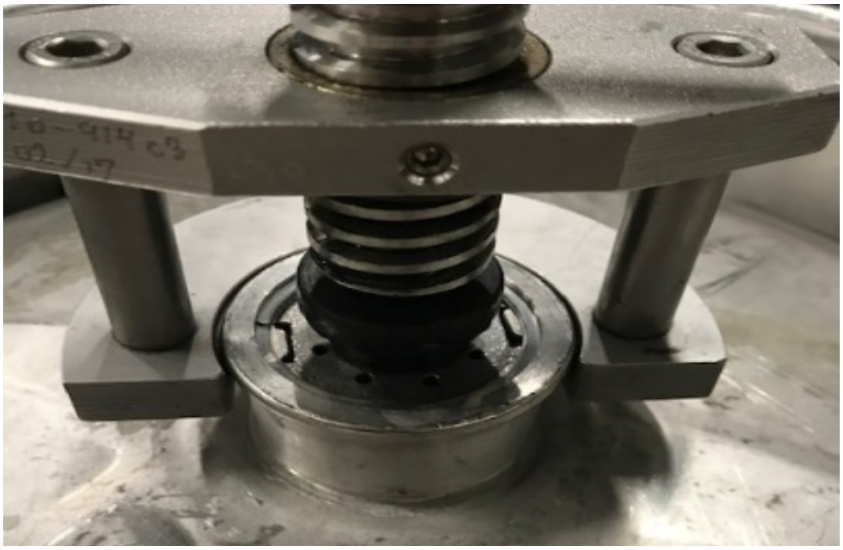
5. To install the locking ring, make sure to start the locking ring end 45 degrees away from the coupler ear channels.

Figure 30: Locking ring installation



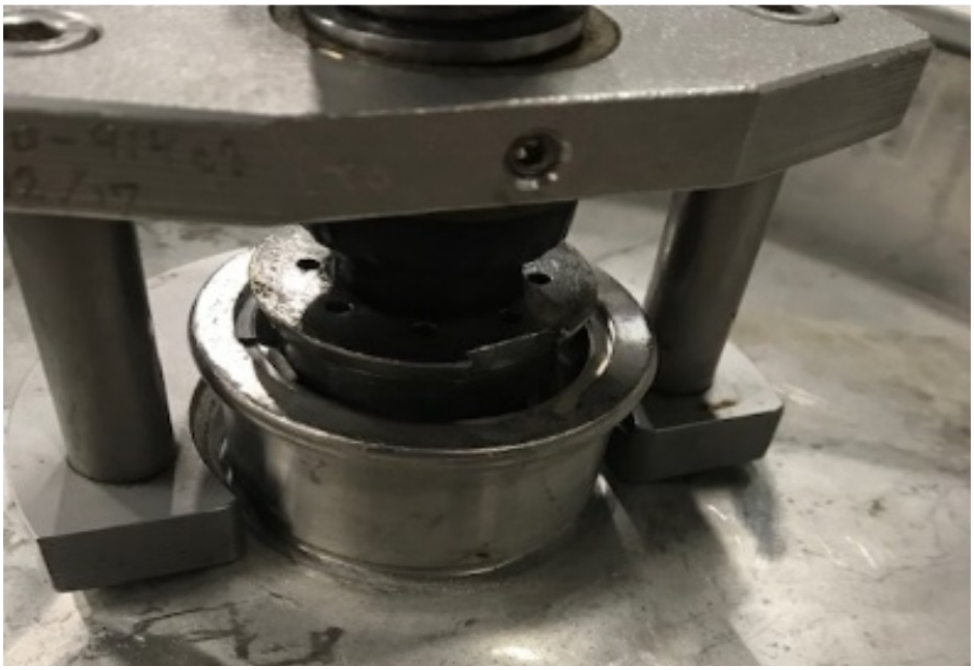
6. Push the locking ring into the channel working counterclockwise.
    6.1 Make sure the lock ring is completely in the channel with equal amounts of the ring showing all the way around.

Figure 31: Locking ring placed in channel



7. Decompress the tool.
    7.1 If the tool does not come out of the coupler easily, the spear will need to be compressed again to fix the locking ring.
    7.2 If the tool can be removed as pictured below, the keg has been rebuilt and can be returned to service.

Figure 32: Rebuilt keg



Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

Not Applicable

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
001	2024/07/12		Kyle Szymanski
002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

Version Approval Number: 002

Approved by: Kyle Szymanski

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